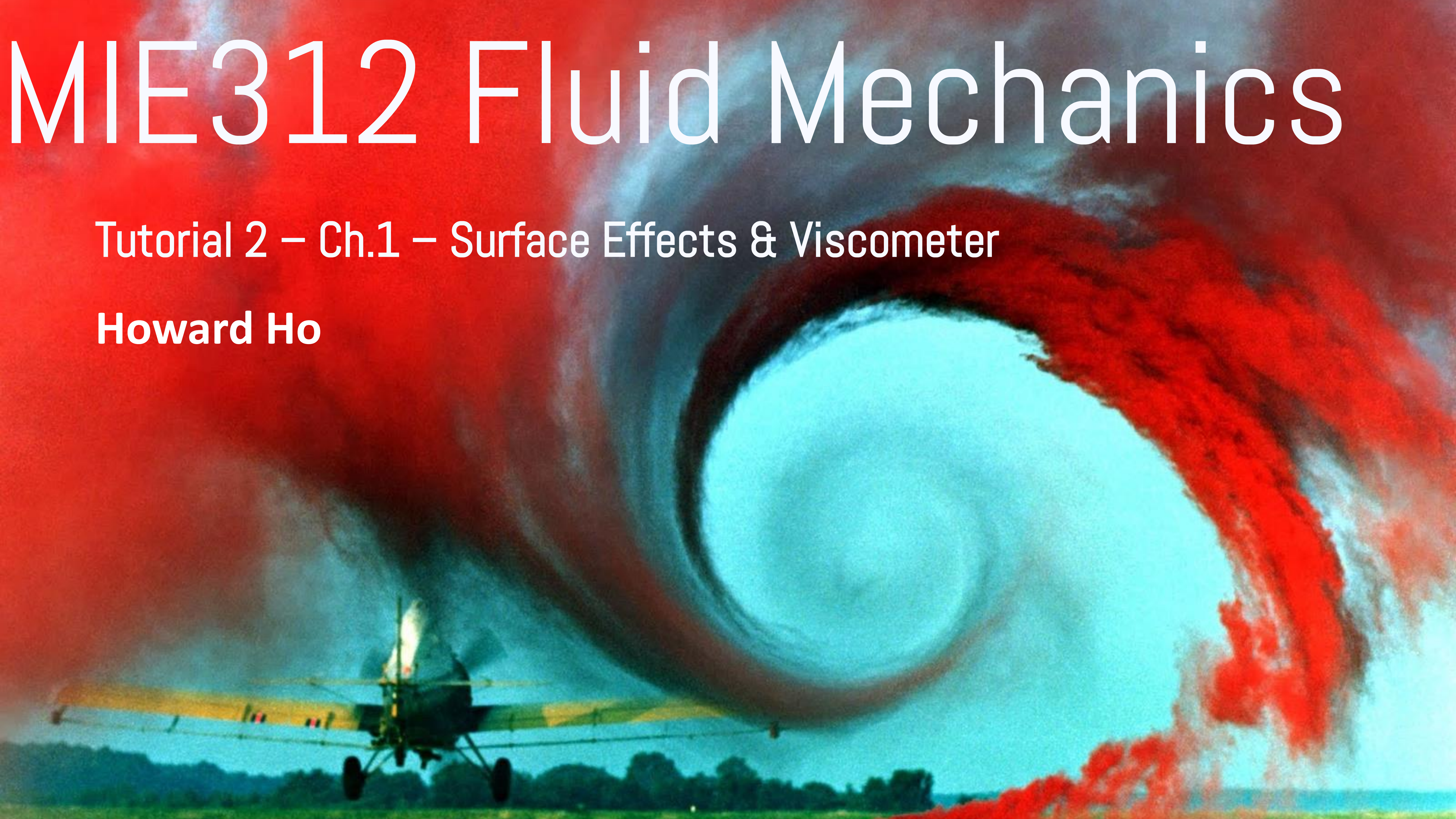
MIE312 Fluid Mechanics

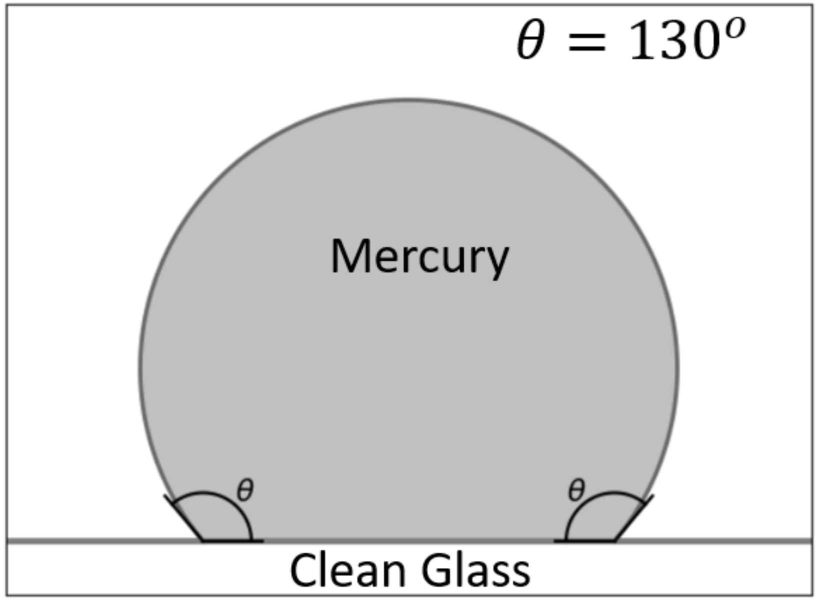
Tutorial 2 – Ch.1 – Surface Effects & Viscometer

Howard Ho



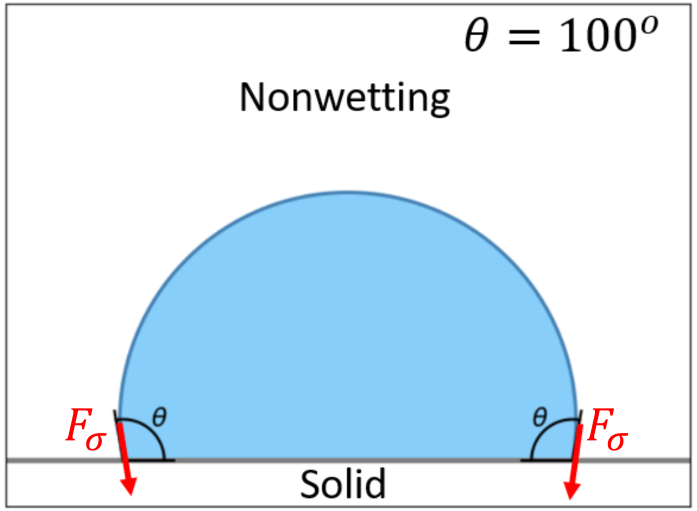
Learning Objective

- Understand the two surface effects of a liquid
  - Surface tension σ
  - Contact angle θ
- Calculate surface tension of droplets and capillary tubes
- Understand how viscometers measure viscosity

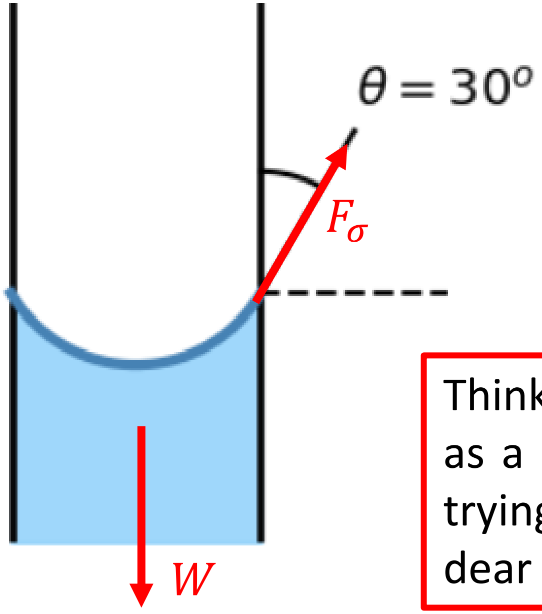


Surface Tension σ

- Not to be confused with Buoyancy (floating is density based)
- Coefficient of surface tension Υ ("upsilon")
  - Unit: N/m or lbf$/ft$
- Cut length dL
- Surface tension force: $F_{\sigma} = dL \times \Upsilon$
- The force acts tangent to the edge of the liquid



σ ("sigma")

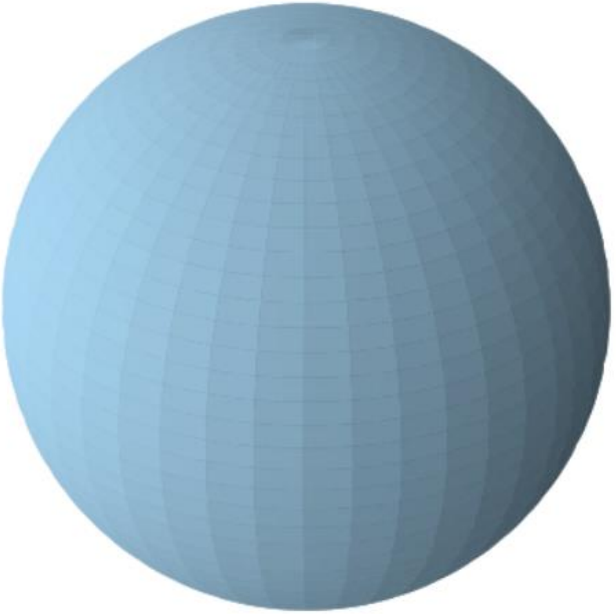


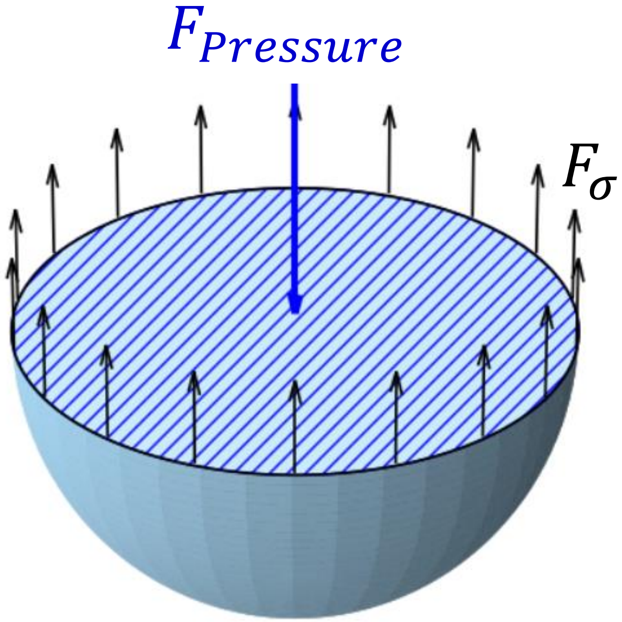
Think of surface tension as a "cheap" plastic bag, trying to hold onto its dear life… 🥲

Water Droplet

$F_\sigma = dL \times \Upsilon$
dL = cut length

- To calculate the surface tension of a water droplet
- We can "pretend to" cut a water droplet in half
- For the water droplet to not expand or contract, the forces must be balance





$$F_{Pressure} = \Delta P \times Area$$
$$F_{Pressure} = \pi R^2 \Delta P$$

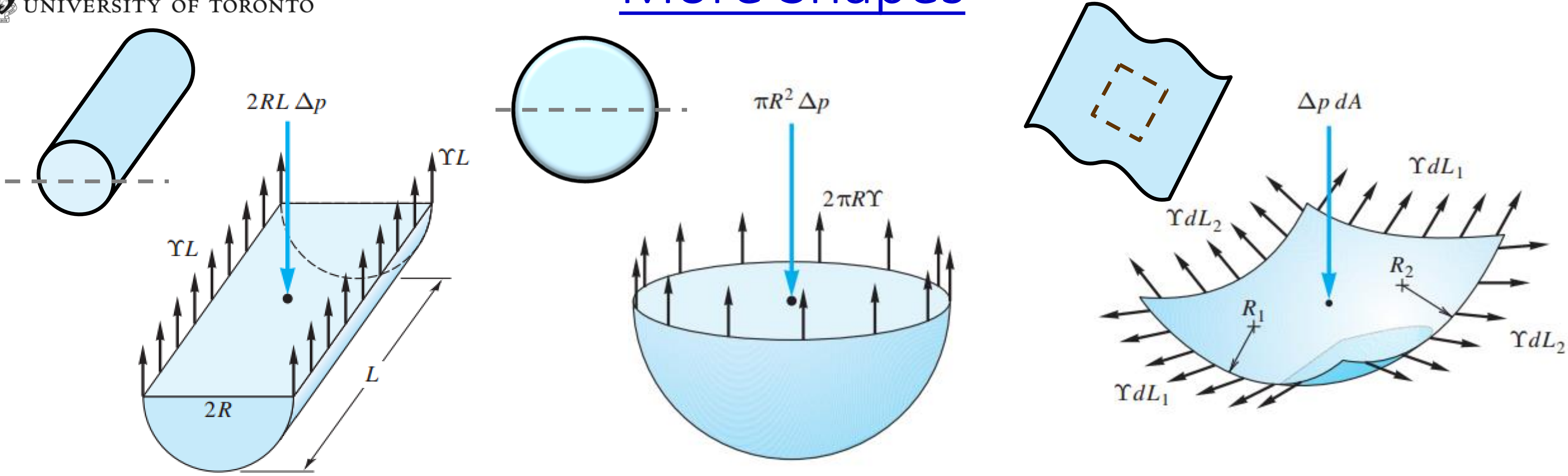
$$F_\sigma = Perimeter \times \Upsilon$$
$$F_\sigma = 2\pi R \Upsilon$$

$$F_{Pressure} = F_\sigma$$
$$\pi R^2 \Delta P = 2\pi R \Upsilon$$
$$R \Delta P = 2\Upsilon$$
$$\Delta P = \frac{2\Upsilon}{R}$$

$$\Delta \mathrm{P} = P_{inside} - P_{outside}$$
e.g.
$$\Delta P = P_{water\ droplet} - P_{air}$$

p.s. we are ignoring gravity here

More Shapes



$$\Delta p = \frac{\Upsilon}{R}$$

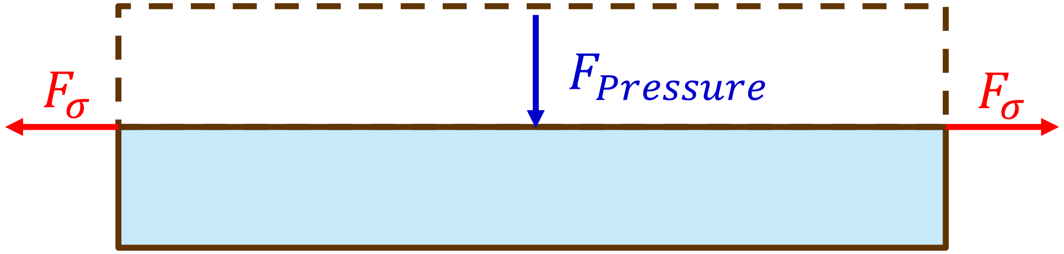
$$\Delta p = \frac{2\Upsilon}{R}$$

$$\Delta p = \Upsilon(R_1^{-1} + R_2^{-1})$$

Fig. 1.11 Pressure change across a curved interface due to surface tension: (*a*) interior of a liquid cylinder; (*b*) interior of a spherical droplet; (*c*) general curved interface.

For geometry (a), we ignore the ends of the cylinder because those surface tension are acting in a different direction than pressure

Side view

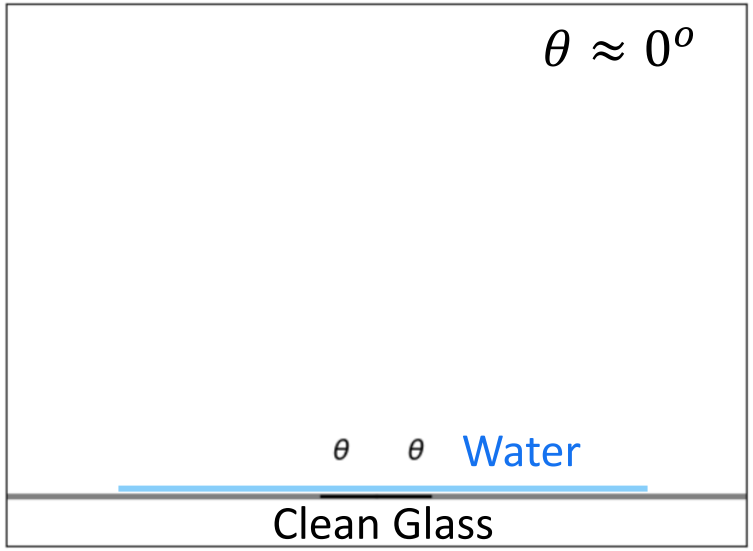


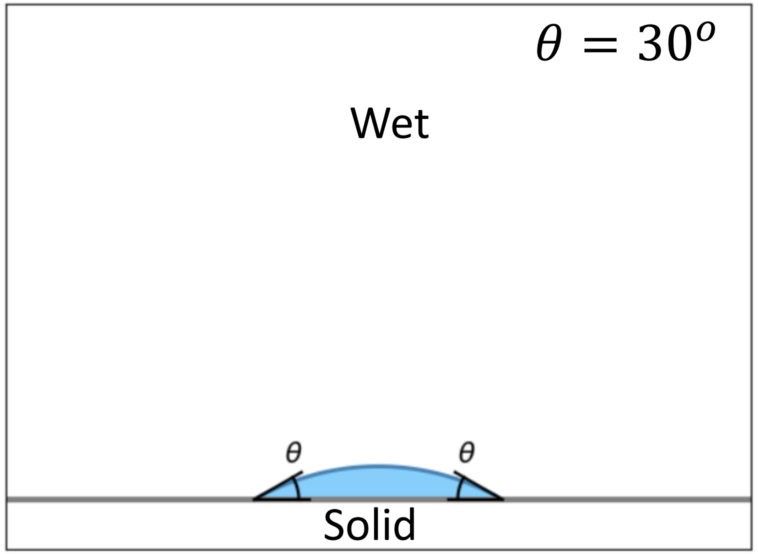
Contact Angle θ

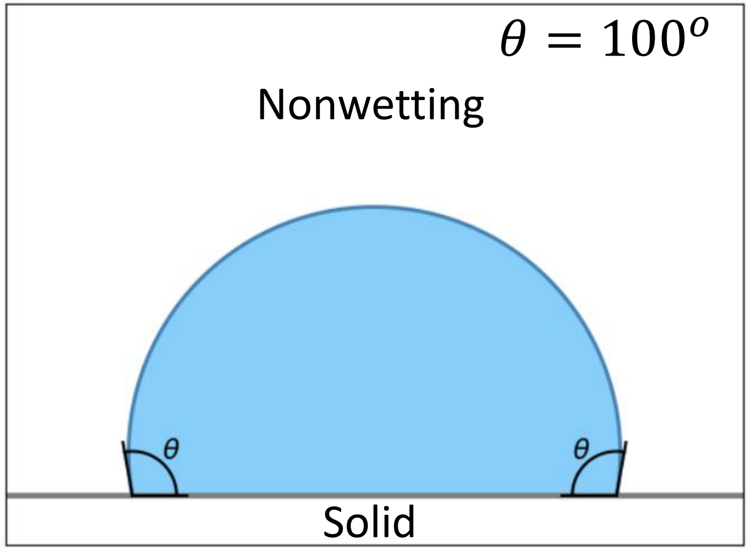
- Contact angle θ is used to describe if the liquid is "wetting" a surface
  - Wet: $\theta < 90^o$
  - Nonwetting: $\theta > 90^o$
- Surface tension coefficient Υ

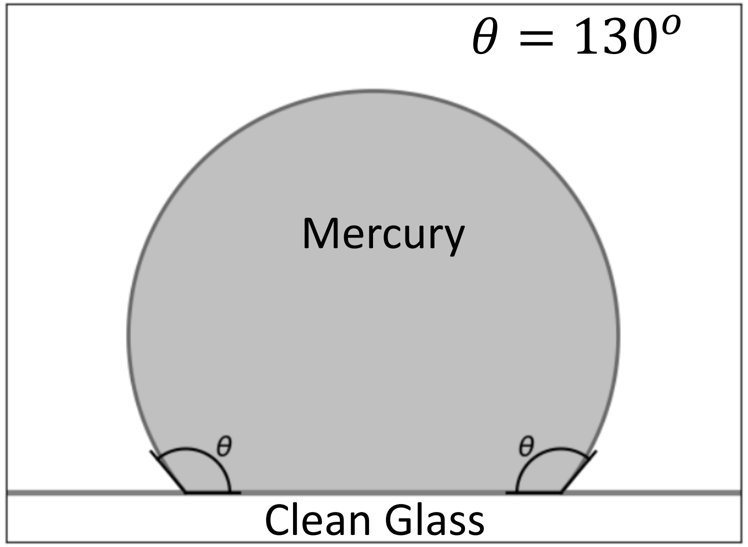
Between medias

$$Y = \begin{cases} 0.0050 \text{ lbf/ft} = 0.073 \text{ N/m} & \text{air–water} \\ 0.033 \text{ lbf/ft} = 0.48 \text{ N/m} & \text{air–mercury} \end{cases} \quad (1.29)$$

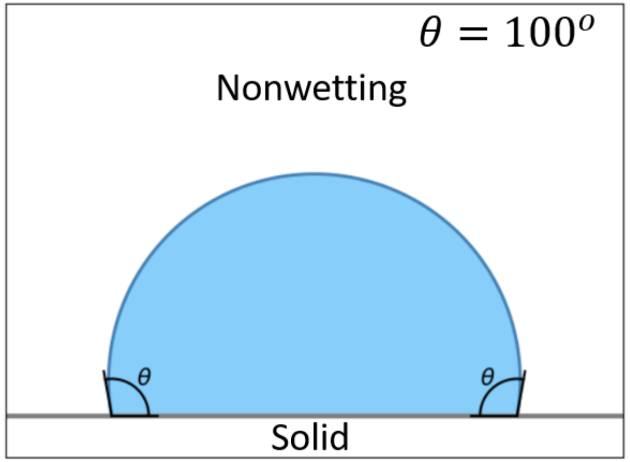








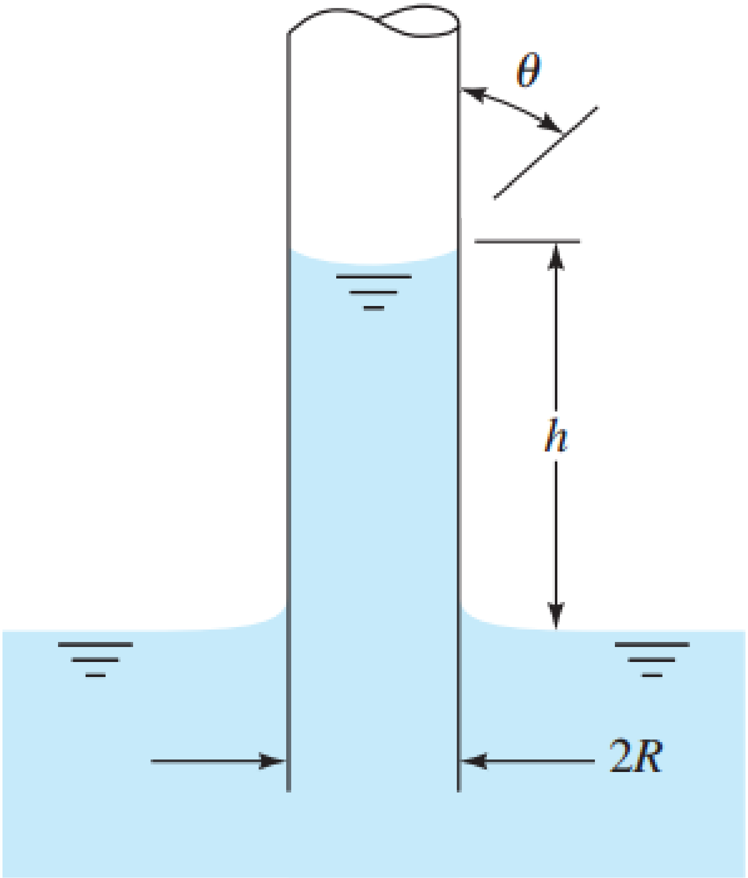
How about vertical tubes?



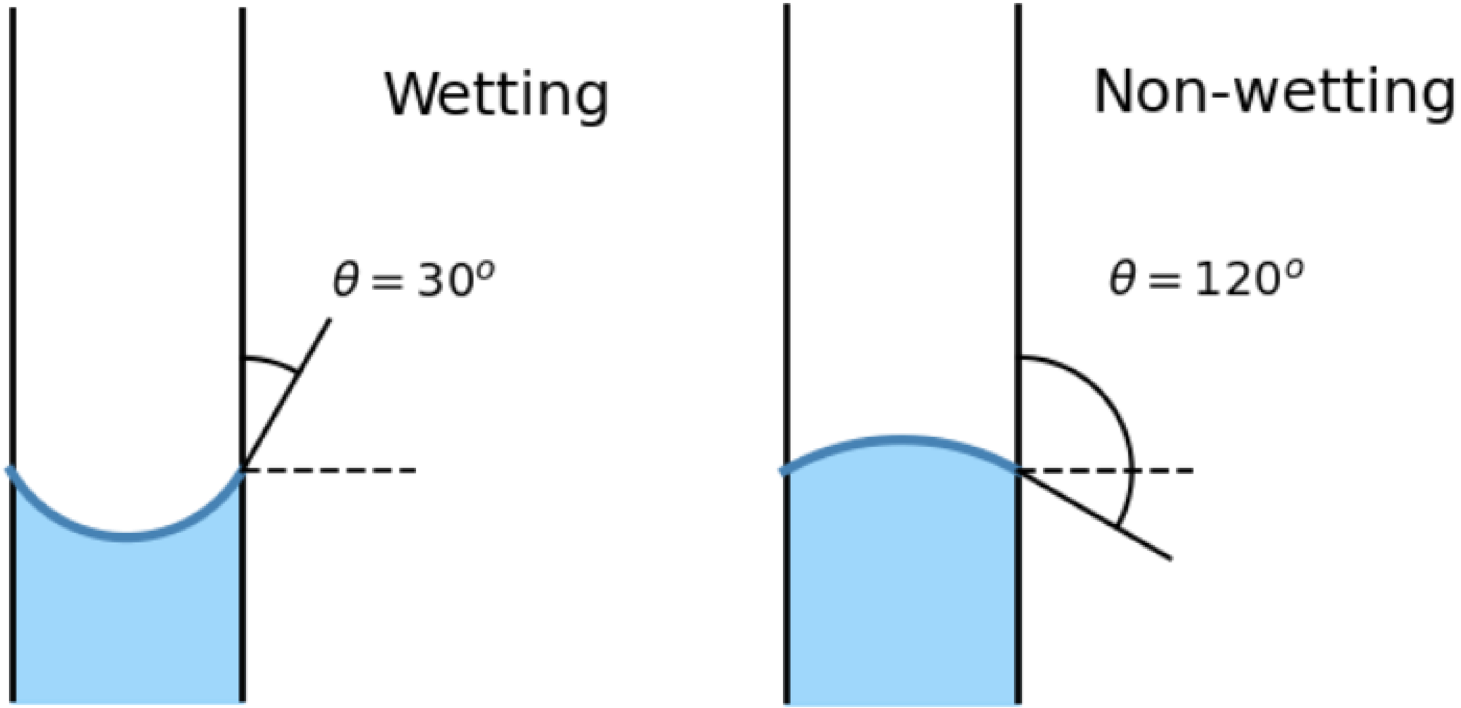
Wet: $\theta < 90^o$

Nonwetting: $\theta > 90^o$

Capillary tubes



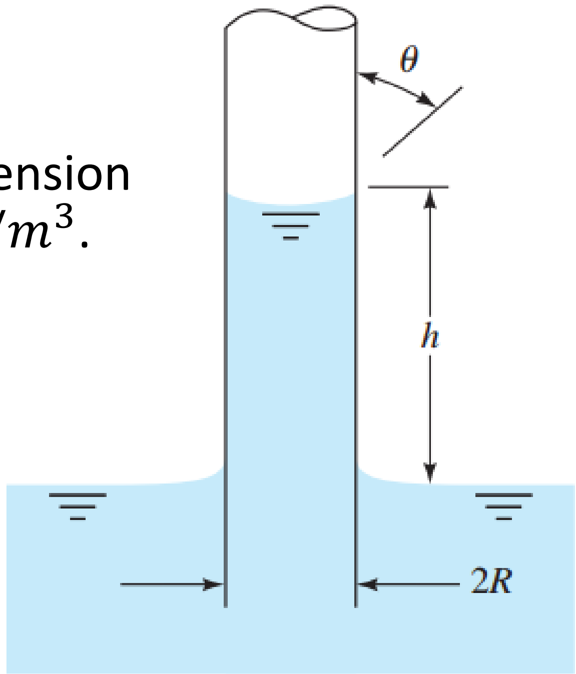
Imagine drawing a line from the edge of water through the tube

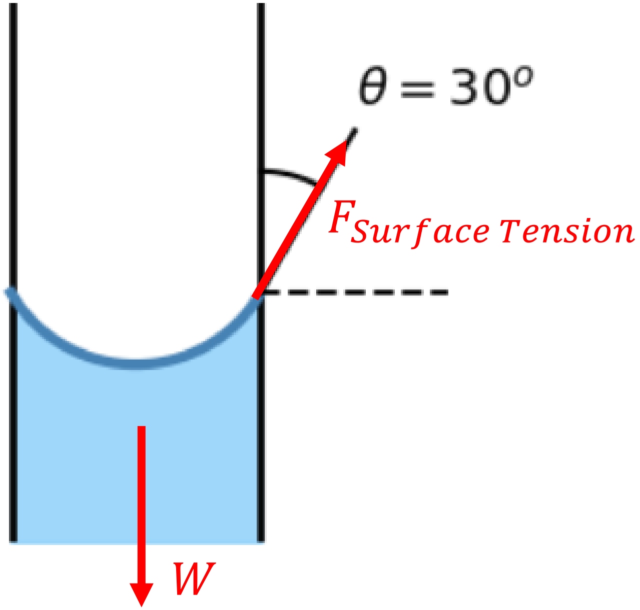


Question Time

Problem 1

Derive an expression for the change in heigh h in a circular tube of a liquid with surface tension Υ and contact angle θ. Then solve for h if $\theta = 130^o, \Upsilon = 0.48\ N/m$, and $\rho = 13{,}600\ kg/m^3$.



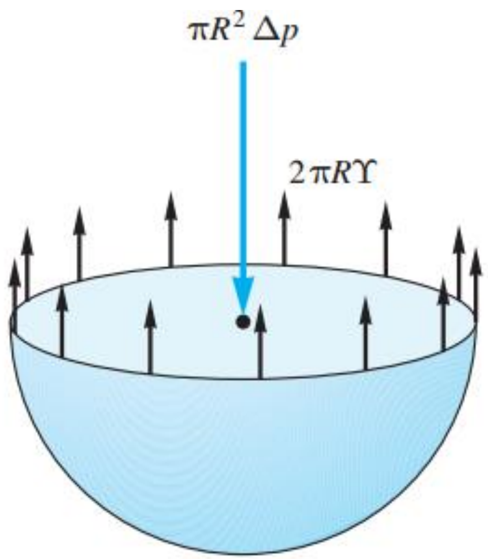


Problem 1

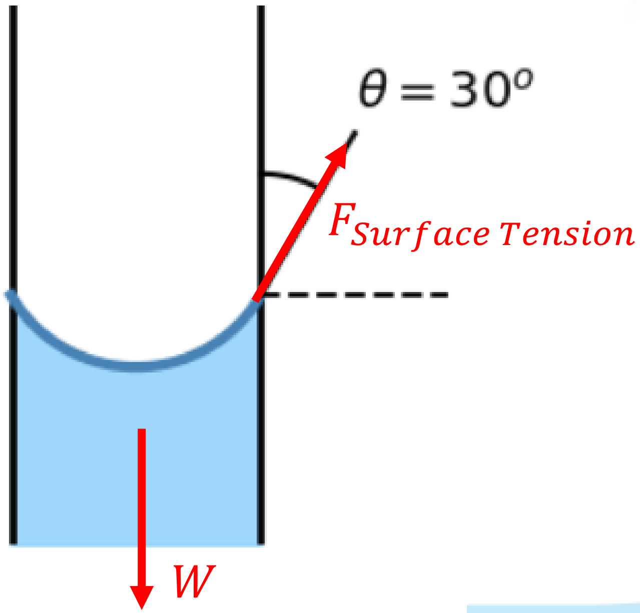
EXAMPLE 1.8

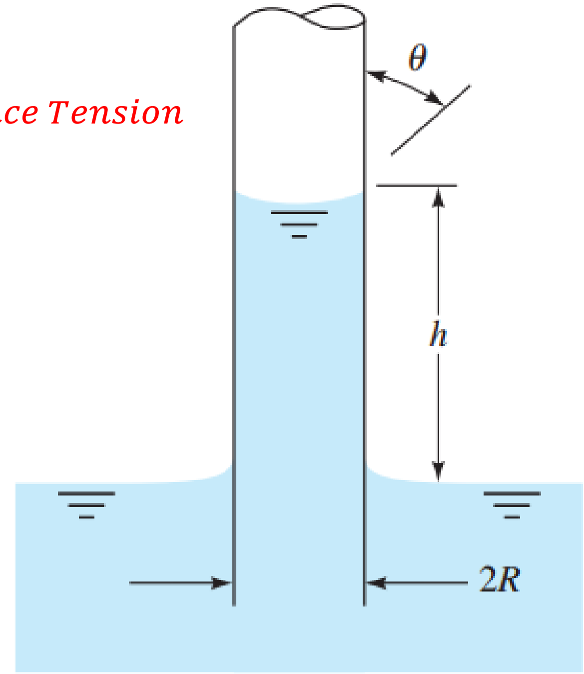
Derive an expression for the change in height h in a circular tube of a liquid with surface tension Υ and contact angle θ, as in Fig. E1.8.

$$\Delta p = \frac{2\Upsilon}{R}$$

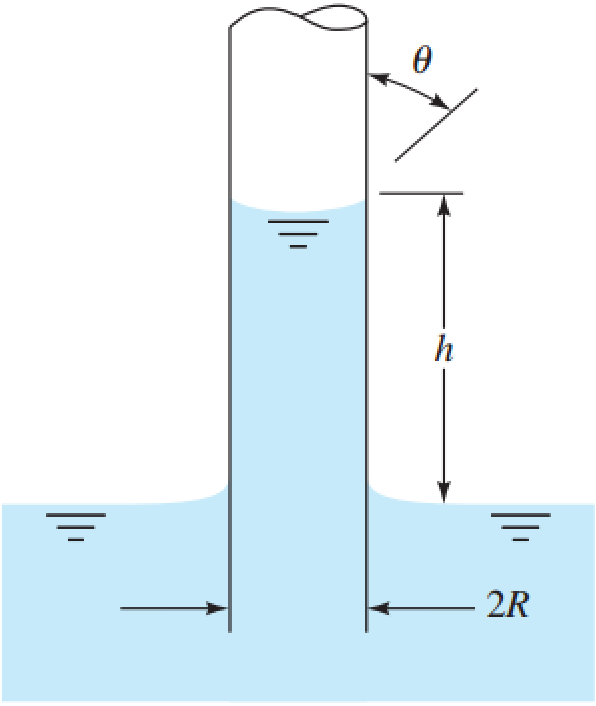


- Pressure force $F_{Pressure} = Pressure \times Area$
- Apply Newton’s 2nd law in vertical direction
- $F_{surface\ tension} \cos\theta = W$
- $2\pi R\Upsilon \cos\theta = \gamma\pi R^2 h$
- $h = \dfrac{2\Upsilon\cos\theta}{\rho g R}$





Specific weight $\gamma = \rho g$ (“gamma”, very extra…)



E1.8

EXAMPLE 1.8

Derive an expression for the change in height h in a circular tube of a liquid with surface tension Υ and contact angle θ, as in Fig. E1.8.

Solution

The vertical component of the ring surface-tension force at the interface in the tube must balance the weight of the column of fluid of height h:

$$2\pi R\Upsilon \cos\theta = \gamma\pi R^2 h$$

Solving for h, we have the desired result:

$$h = \frac{2\Upsilon \cos\theta}{\gamma R} \qquad \textit{Ans.}$$

Thus the capillary height increases inversely with tube radius R and is positive if $\theta < 90°$ (wetting liquid) and negative (capillary depression) if $\theta > 90°$.

Suppose that $R = 1$ mm. Then the capillary rise for a water–air–glass interface, $\theta \approx 0°$, $\Upsilon = 0.073$ N/m, and $\rho = 1000$ kg/m^3 is

$$h = \frac{2(0.073\ \text{N/m})(\cos 0°)}{(1000\ \text{kg/m}^3)(9.81\ \text{m/s}^2)(0.001\ \text{m})} = 0.015\ (\text{N}\cdot\text{s}^2)/\text{kg} = 0.015\ \text{m} = 1.5\ \text{cm}$$

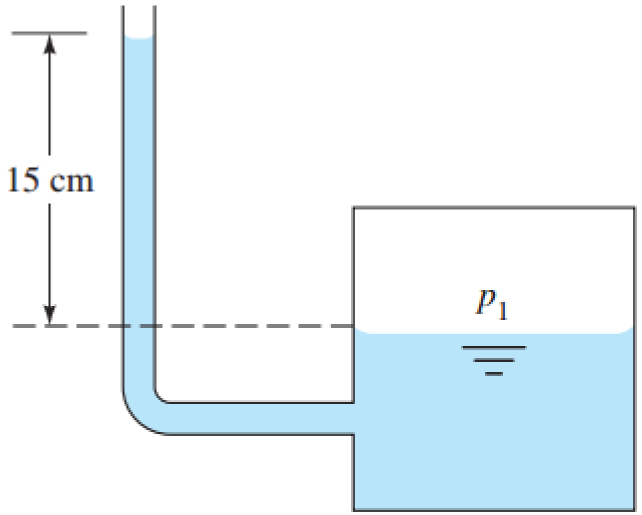
For a mercury–air–glass interface, with $\theta = 130°$, $\Upsilon = 0.48$ N/m, and $\rho = 13{,}600$ kg/m^3, the capillary rise is

$$h = \frac{2(0.48)(\cos 130°)}{13{,}600(9.81)(0.001)} = -0.0046\ \text{m} = -0.46\ \text{cm}$$

When a small-diameter tube is used to make pressure measurements (Chap. 2), these capillary effects must be corrected for.

Problem 2*

The system in Fig. P1.65 is used to estimate the pressure p_1 in the tank by measuring the 15-cm height of liquid in the 1-mm-diameter tube. The fluid is at 60°C. Calculate the true fluid height in the tube and the percent error due to capillarity if the fluid is (a) water; and (b) mercury.



Solution: This is a somewhat more realistic variation of Ex. 1.9. Use values from that example for contact angle θ:

(a) Water at 60°C: $\gamma \approx 9640\ \text{N/m}^3$, $\theta \approx 0°$:

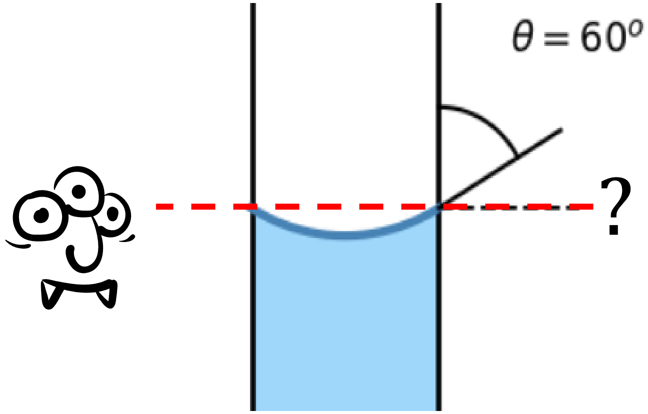
$$h = \frac{4Y\cos\theta}{\gamma D} = \frac{4(0.0662\ \text{N/m})\cos(0°)}{(9640\ \text{N/m}^3)(0.001\ \text{m})} = 0.0275\ \text{m},$$

or: $\Delta h\text{true} = 15.0 - 2.75\ \text{cm} \approx \mathbf{12.25\ cm\ (+22\%\ error)}$ *Ans.* (a)

(b) Mercury at 60°C: $\gamma \approx 132200\ \text{N/m}^3$, $\theta \approx 130°$:

$$h = \frac{4Y\cos\theta}{\gamma D} = \frac{4(0.47\ \text{N/m})\cos 130°}{(132200\ \text{N/m}^3)(0.001\ \text{m})} = -0.0091\ \text{m},$$

or: $\Delta h_{true} = 15.0 + 0.91 \approx \mathbf{15.91\ cm\ (-6\%\ error)}$ *Ans.* (b)



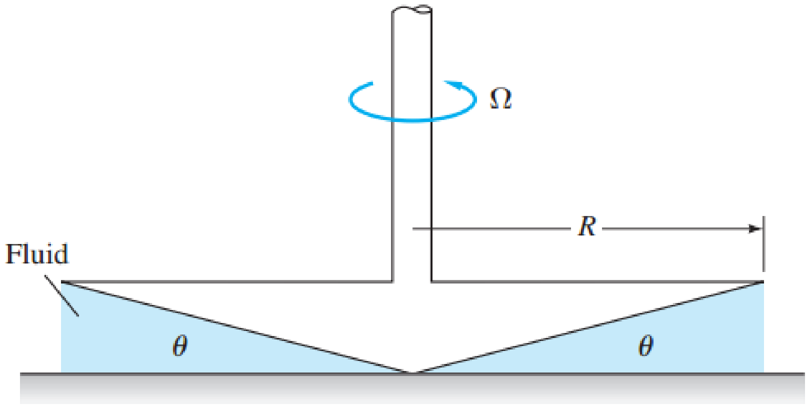
Equation for error:

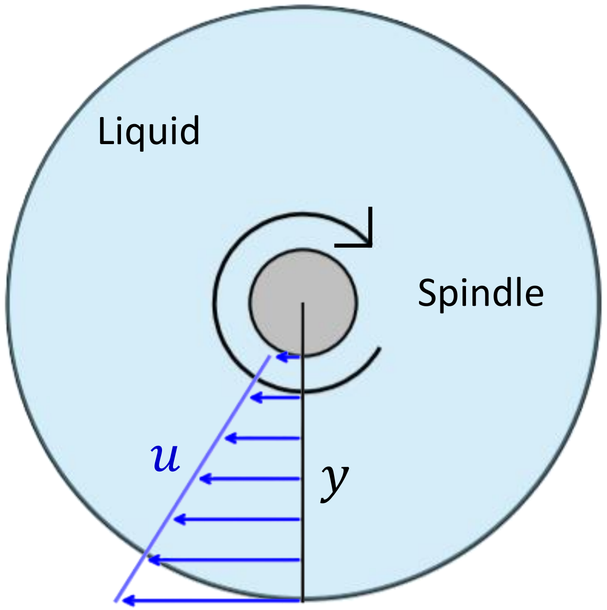
$$\%\ error = \frac{Measure - True}{True} \times 100\%$$

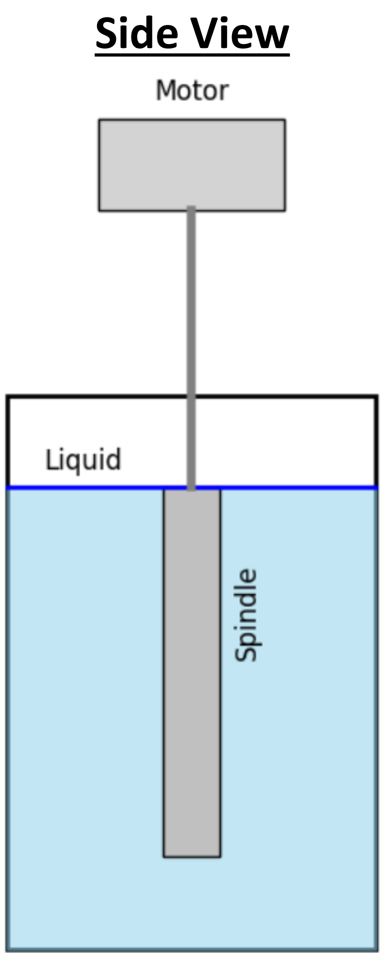
The only difference here compared with Problem 1 is P_1 is enclosed, and they used diameter D for the tube instead of radius, that's why the h equation starts with 4 instead of 2 ($D = 2R$).

Rotating Viscometer

- Used to measure viscosity of fluids
- Shear stress τ is generated from the rotating spindle
- Recall $\tau = \mu \frac{du}{dy}$
- Torque from the shear stress is measure by the motor
- More torque = More viscous
- They come in different shapes





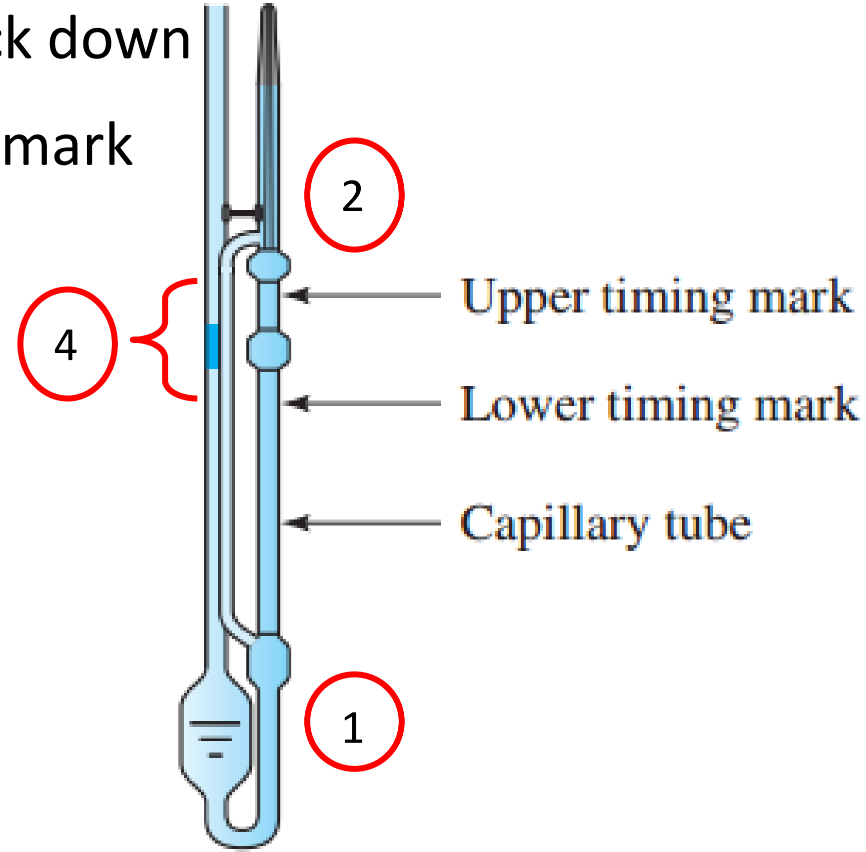


Capillary Viscometer*

Another way of measuring viscosity

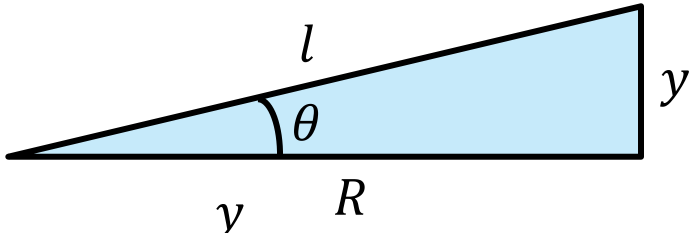
1. Testing liquid is first stored in the lower chamber
2. Liquid then suction pumped to the upper timing marker
3. Suction pump is then released, allowing liquid to flow back down
4. Time takes for liquid to travel from upper to lower timing mark measured

- Viscosity of liquid is then calculated
- Longer it takes = more viscous



Problem 3

The image display a cone-plate viscometer. The angle of the cone is very small, so that sin θ < θ , and the gap is filled with the test liquid. The torque M to rotate the cone at a rate ω is measured. Assuming a linear velocity profile in the fluid film, derive an expression for the torque M measure as a function of (μ, R, ω, θ)



$$\tan\theta = \frac{y}{R}$$

$$y = R\tan\theta$$

$$u_{max} = R\omega$$

Linear velocity profile

$$\frac{du}{dy} = \frac{R\omega}{R\tan\theta}$$

$$\tau = \mu\frac{du}{dy}$$

$$\tau = \frac{\mu\omega}{\tan\theta}$$

τ is same everywhere (linear profile in fluid film)

$$F_{shear} = \tau \times Area$$

$$dF_{shear} = \tau\, dA$$

$$dM = r\, dF_{shear}$$

$$dM = r\left(\frac{\mu\omega}{\tan\theta}\right)\left(\frac{2\pi r}{\cos\theta}\right)dr$$

$$\int dM = \int_0^R \frac{2\mu\pi\omega}{\sin\theta} r^2 dr$$

$$M = \frac{2\mu\pi\omega}{\sin\theta}\frac{R^3}{3}$$

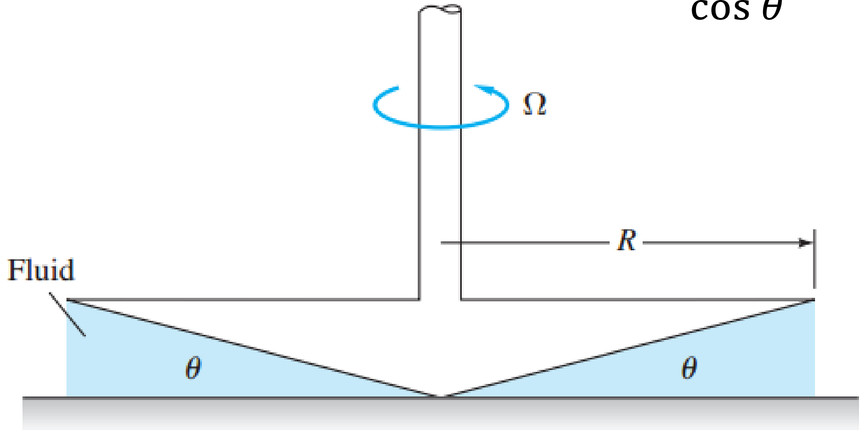
$$A_{cone} = \pi R l$$

$$\cos\theta = \frac{R}{l}$$

$$l = \frac{R}{\cos\theta}$$

$$A_{cone} = \frac{\pi R^2}{\cos\theta}$$

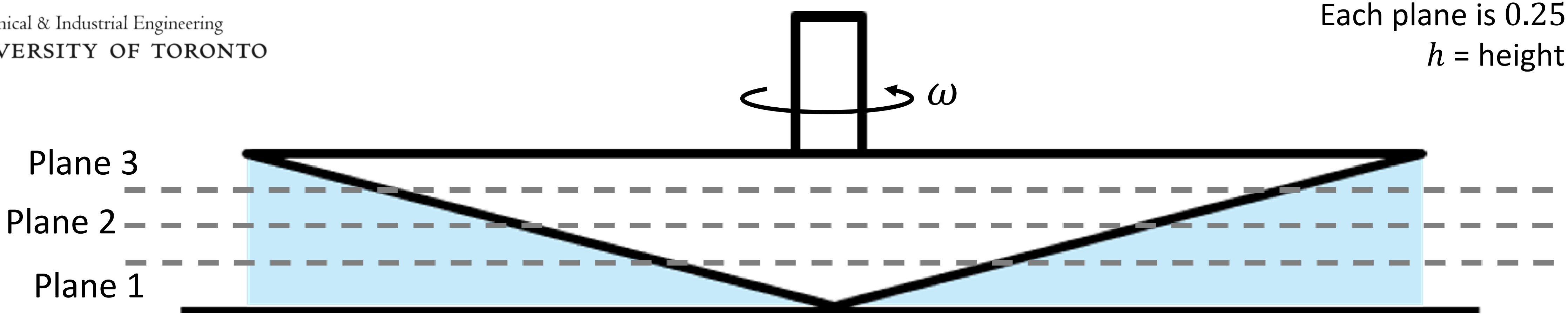
Hint: $dA = \dfrac{2\pi r}{\cos\theta} dr$



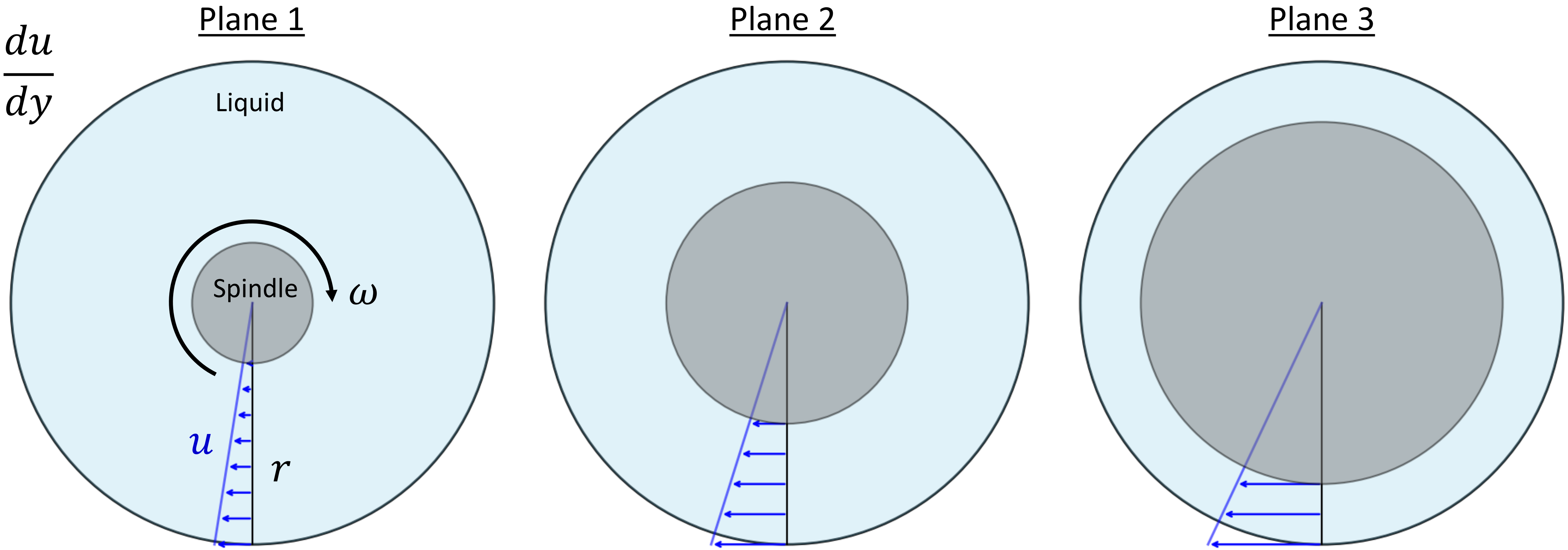
P1.56

Ω is same as ω here, we tend to use lower case ω in fluids

Each plane is $0.25\ h$ apart
h = height of cone



$$\tau = \mu \frac{du}{dy}$$



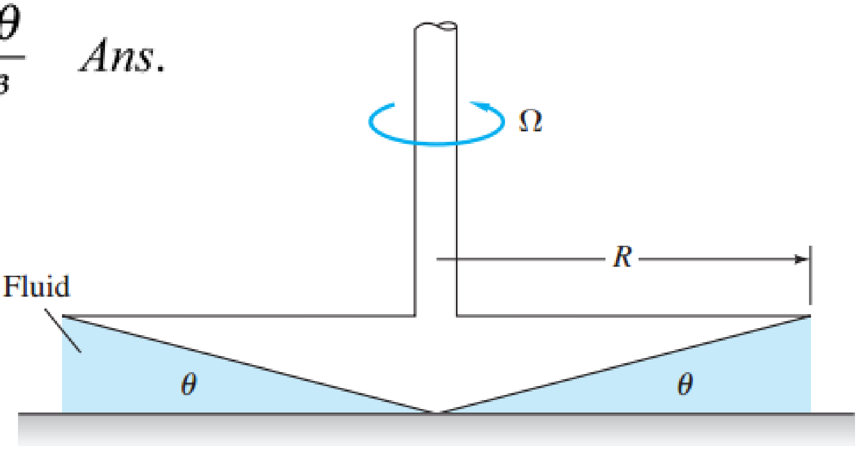
Problem 3

The device in Fig. P1.56 is called a cone-plate viscometer. The angle of the cone is very small, so that sin θ < θ , and the gap is filled with the test liquid. The torque M to rotate the cone at a rate ω is measured. Assuming a linear velocity profile in the fluid film, derive an expression for fluid viscosity μ as a function of (M , R , ω , θ).

Solution: For any radius $r \le R$, the liquid gap is $h = r\tan\theta$. Then

$$d(\text{Torque}) = dM = \tau\, dA_w r = \left(\mu \frac{\Omega r}{r\tan\theta}\right)\left(2\pi r \frac{dr}{\cos\theta}\right) r, \quad \text{or}$$

$$M = \frac{2\pi\Omega\mu}{\sin\theta}\int_0^R r^2\, dr = \frac{2\pi\Omega\mu R^3}{3\sin\theta}, \quad \text{or:} \quad \mu = \frac{3M\sin\theta}{2\pi\Omega R^3} \quad \textit{Ans.}$$



P1.56

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Link to a pretty cool video

https://www.youtube.com/watch?v=E2unnSK7WTE

This Is Why Water Striders Make Terrible Lifeguards | Deep Look